

```
jane@Janes-MacBook-Air 3tier-terraform % terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.100.0
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
jane@Janes-MacBook-Air 3tier-terraform % terraform plan
module.core.aws_eip.nat: Refreshing state... [id=eipalloc-017a583d55633eb09]
module.core.aws_vpc.main: Refreshing state... [id=vpc-0caea84e0341ede90]
module.core.aws_internet_gateway.igw: Refreshing state... [id=igw-0f9dd53e4ba69cd34]
module.core.aws_subnet.database[1]: Refreshing state... [id=subnet-008e4972495ac252c]
module.core.aws_subnet.database[0]: Refreshing state... [id=subnet-07433778637ae8fc9]
```

```
jane@Janes-MacBook-Air 3tier-terraform % terraform apply
module.core.aws_eip.nat: Refreshing state... [id=eipalloc-017a583d55633eb09]
module.core.aws_vpc.main: Refreshing state... [id=vpc-0caea84e0341ede90]
module.core.aws_internet_gateway.igw: Refreshing state... [id=igw-0f9dd53e4ba69cd34]
module.core.aws_subnet.database[0]: Refreshing state... [id=subnet-07433778637ae8fc9]
module.core.aws_subnet.public[1]: Refreshing state... [id=subnet-0cffbd587eda26c9a]
module.core.aws_subnet.public[0]: Refreshing state... [id=subnet-0992d4b0cbe8d4bc6]
module.core.aws_security_group.alb_sg: Refreshing state... [id=sg-017837756c324da96]
module.core.aws_subnet.database[1]: Refreshing state... [id=subnet-008e4972495ac252c]
module.core.aws_subnet.private[0]: Refreshing state... [id=subnet-06b504ca38affd133]
module.core.aws_subnet.private[1]: Refreshing state... [id=subnet-00226ab3e66de2c6d]
module.compute.aws_lb_target_group.app_tg: Refreshing state... [id=arn:aws:elasticloadbal
module.core.aws_route_table.public: Refreshing state... [id=rtb-03cf486b679563d6b]
```

Outputs:

```
alb_dns_name = "3tier-public-alb-1906925261.us-east-1.elb.amazonaws.com"
db_endpoint  = "app-mysql-db.citsuoc4ck05.us-east-1.rds.amazonaws.com:3306"
db_subnet_ids = [
  "subnet-07433778637ae8fc9",
  "subnet-008e4972495ac252c",
]
private_subnet_ids = [
  "subnet-06b504ca38affd133",
  "subnet-00226ab3e66de2c6d",
]
public_subnet_ids = [
  "subnet-0992d4b0cbe8d4bc6",
  "subnet-0cffbd587eda26c9a",
]
vpc_id = "vpc-0caea84e0341ede90"
```

Set Up the VPC

Your VPCs

VPCs | VPC encryption controls

Your VPCs (2) Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6
☐	3tier-vpc	vpc-0caea84e0341ede90	Available	-	-	Off	10.0.0.0/16	-

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	3tier-public-subnet-2	subnet-0c7fbd587eda26c9a	Available	vpc-0caea84e0341ede90 3tier...	Off	10.0.2.0/24	-
☐	3tier-public-subnet-1	subnet-0992d4b0cbe8d4bc6	Available	vpc-0caea84e0341ede90 3tier...	Off	10.0.1.0/24	-
☐	3tier-private-subnet-2	subnet-00226ab3e65de2c6d	Available	vpc-0caea84e0341ede90 3tier...	Off	10.0.4.0/24	-
☐	3tier-private-subnet-1	subnet-06b504ca38affd133	Available	vpc-0caea84e0341ede90 3tier...	Off	10.0.3.0/24	-
☐	3tier-db-subnet-2	subnet-008e4972495ac252c	Available	vpc-0caea84e0341ede90 3tier...	Off	10.0.6.0/24	-
☐	3tier-db-subnet-1	subnet-07433778637ae8fc9	Available	vpc-0caea84e0341ede90 3tier...	Off	10.0.5.0/24	-

Route tables (4) Info

Last updated 9 minutes ago

Actions

Create route table

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	3tier-public-rt	rtb-03cf486b679563d6b	2 subnets	-	No	vpc-0caea84e0341ede90 3tier...	204312534875
☐	3tier-private-rt	rtb-0bd20063c79a7cc30	4 subnets	-	No	vpc-0caea84e0341ede90 3tier...	204312534875

igw-0f9dd53e4ba69cd34 / 3tier-igw

Details Info

Internet gateway ID
igw-0f9dd53e4ba69cd34

State
Attached

VPC ID
vpc-0caea84e0341ede90 | 3tier-vpc

Owner
204312534875

Tags (1)

Search tags

Key	Value
Name	3tier-igw

nat-0de91c742cedf9b03 / 3tier-nat-gateway

Actions

Details

NAT gateway ID
nat-0de91c742cedf9b03

NAT gateway ARN
arn:aws:ec2:us-east-1:204312534875:natgateway/nat-0de91c742cedf9b03

VPC
vpc-0caea84e0341ede90 | 3tier-vpc

Connectivity type
Public

Primary public IPv4 address
100.49.182.145

Subnet
subnet-0992d4b0cbe8d4bc6 | 3tier-public-subnet-1

State
Available

Primary private IPv4 address
10.0.1.28

Created
Friday, March 20, 2026 at 21:14:06 PDT

State message Info
-

Primary network interface ID
eni-0ecc9a22ffd393bf9

Deleted
-

Secondary IPv4 addresses

Monitoring

Tags

Secondary IPv4 addresses

Edit secondary IPv4 address associations

Search

Private IPv4 address	Network interface ID	Status	Failure message
----------------------	----------------------	--------	-----------------

Secondary IPv4 addresses are not available for this nat gateway.

Deploy the Frontend (Tier 1)

3tier-public-alb

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

Active

Hosted zone

Z35SXDOTRQ7X7K

VPC

vpc-0caea84e0341ede90

Availability Zones

subnet-0992d4b0cbe8d4bc6 us-east-1a (use1-az1)
subnet-0cfffbd587eda26c9a us-east-1b (use1-az2)

Load balancer IP address type

IPv4

Date created

March 20, 2026, 21:14 (UTC-07:00)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:204312534875:loadbalancer/app/3tier-public-alb/0d6a7527446d68e3

DNS name

3tier-public-alb-1906925261.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1)

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 >

Protocol:Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

mTLS

Trust store

Forward to target group

3tier-app-tg: 1 (100%)

Target group stickiness: Off

HTTP:80

1 rule

ARN

Not applicable

Not applicable

Not applicable

Not applicable

3-tier app running

EC2 is in private subnet behind ALB.

Deploy the Backend (Tier 2)

Instance details

Instance summary for i-0eac5a68be98132bf (3tier-app-instance)

Updated less than a minute ago

Connect

Instance state

Actions

Instance ID

i-0eac5a68be98132bf

IPv6 address

-

Hostname type

IP name: ip-10-0-3-247.ec2.internal

Answer private resource DNS name

-

Auto-assigned IP address

-

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-3-247.ec2.internal

Instance type

t3.micro

VPC ID

vpc-0caea84e0341ede90 (3tier-vpc)

Subnet ID

subnet-06b504ca38affd133 (3tier-private-subnet-1)

Instance ARN

arn:aws:ec2:us-east-1:204312534875:instance/i-0eac5a68be98132bf

Private IPv4 addresses

10.0.3.247

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

3tier-app-asg

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details

AMI ID

ami-02dfbd4ff395f2a1b

Monitoring

disabled

Platform details

Linux/UNIX

AMI name

...

Allowed image

...

Termination protection

...

Details

Target group details

lancing:us-east-1:204312534875:targetgroup/3tier-app-tg/9b24e7faea5a0ae0

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

[vpc-0caea84e0341ede90](#)

IP address type

IPv4

Load balancer

[3tier-public-alb](#)

1

Total targets

1

Healthy

0

Anomalous

0

Unhealthy

0

Unused

0

Initial

0

Draining

► Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (1)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Q Filter targets

< 1 >

☐

Instance ID

Name

Port

Zone

Health status

Health status details

Administrative over...

Override details

Launch...

Anom...

☐

[i-0eac5a68be98132bf](#)

3tier-app-insta...

80

us-east-1a (us...

Healthy

-

No override

No override is currently...

March 20,...

Healthy

Deploy the Database (Tier 3)

Aurora and RDS

> Events

Aurora and RDS

Dashboard

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Events (2)

Q Filter by event

< 1 >

Source

Type

Time

Message

[app-mysql-db](#)

Instances

March 20, 2026, 21:30 (UTC-07:00)

DB instance created

[app-mysql-db](#)

Instances

March 20, 2026, 21:29 (UTC-07:00)

DB instance restarted

app-mysql-db

Modify

Actions

Summary

DB identifier

app-mysql-db

CPU

4.06%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

us-east-1a

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Recommendations

Database Insights is now available

Database Insights combines database metrics and logs into a unified view in CloudWatch to speed up database troubleshooting. For enhanced monitoring capabilities and advanced analytics, modify your database to use the Advanced mode of Database Insights.

Modify

CloudWatch (24)

Search by metric

Alarm recommendations

Investigate with AI - new

1h

3h

12h

1d

3d

1w

Custom

UTC timezone

Explore related

BinLogDiskUsage

Bytes

app-mysql-db

CPUCreditBalance

Count

app-mysql-db

CPUCreditUsage

Count

app-mysql-db

CPUUsage

Percent

app-mysql-db

CPUUtilization

Percent

app-mysql-db

CPUUsage

Percent

app-mysql-db

CPUUsage

Percent

app-mysql-db

Secure the Architecture

sg-0511a8a1f902ceb62 - 3tier-ec2-sg

Details

Security group name

3tier-ec2-sg

Security group ID

sg-0511a8a1f902ceb62

Description

Allow HTTP from ALB to EC2

VPC ID

vpc-0caea84e0341ede90

Owner

204312534875

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Q Search

Manage tags

Edit inbound

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0f1709b3d5de3bde2	-	HTTP	TCP	80	sg-017837756c3

sg-0187f1a08933e8745 - 3tier-db-sg

Details

Security group name

3tier-db-sg

Security group ID

sg-0187f1a08933e8745

Description

Allow MySQL from EC2

VPC ID

vpc-0caea84e0341ede90

Owner

204312534875

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

sg-017837756c324da96 - 3tier-alb-sg

Details

Security group name

3tier-alb-sg

Security group ID

sg-017837756c324da96

Description

Allow HTTP from internet to ALB

VPC ID

vpc-0caea84e0341ede90

Owner

204312534875

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Q Search

Manage tags

Edit inbound

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-09db232f9115a84b7	IPv4	HTTP	TCP	80	0.0.0.0/0